Covert Pictorial Narrative and Multimedia Content Generation in Image Communication Networks

Fernando May

Instituto Politécnico Nacional, Mexico

Motivation

- Image communication channels are massive, largely unexamined covert vectors
- Three fundamental requirements: capacity, imperceptibility, security
- Fourth dimension: narrative coherence (hidden payload carries structure)
- Need unified benchmark across spatial/transform/learned/adaptive paradigms

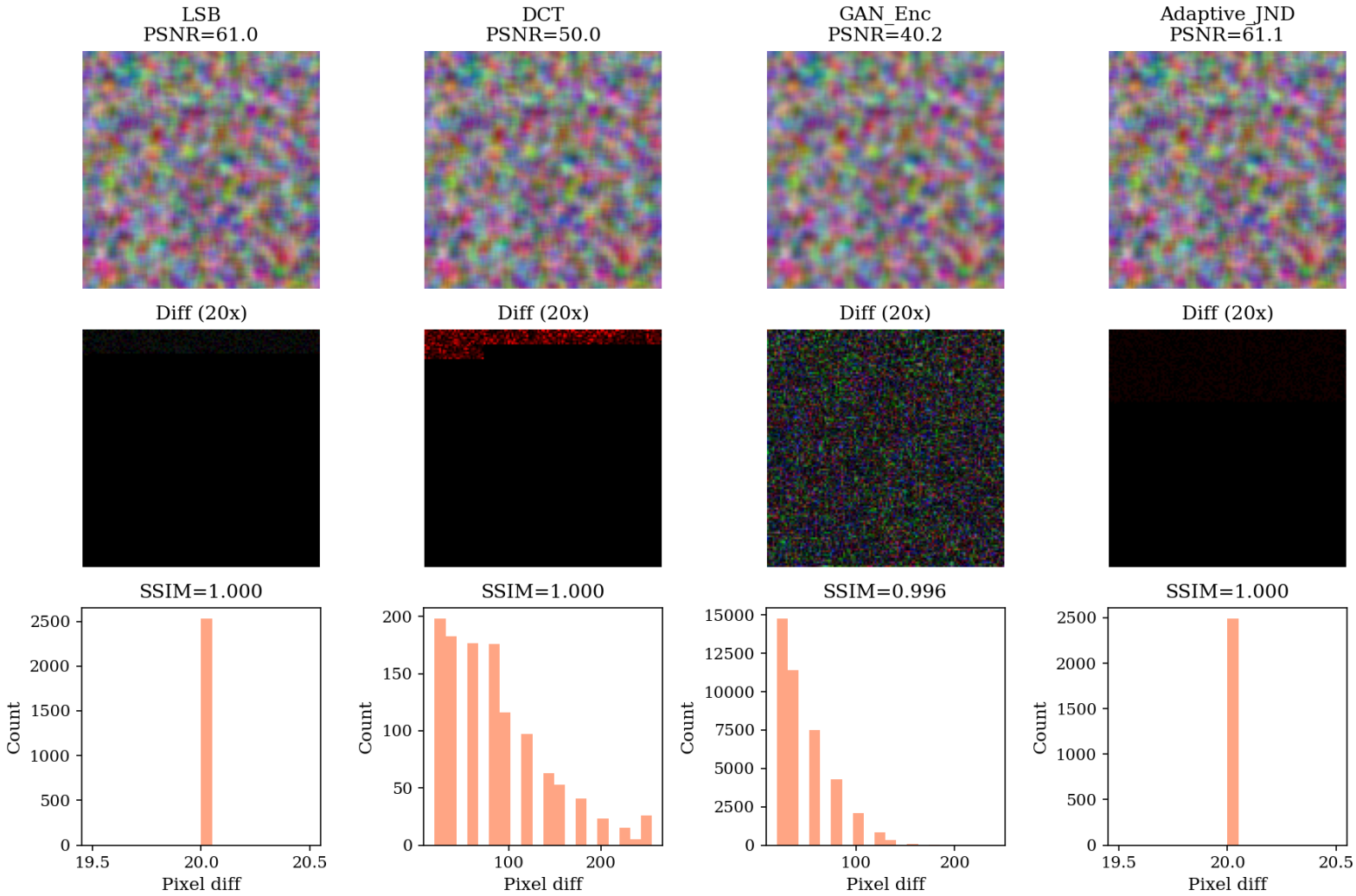
Steganographic Methods

1. **LSB Substitution** – k bits per pixel in spatial domain (baseline)
2. **DCT Modulation** (F5 variant) – parity of mid-frequency coefficients
3. **GAN-inspired Encoder** – SVD latent space projection + perturbation
4. **Adaptive JND** – embedding strength $\propto$ local JND threshold

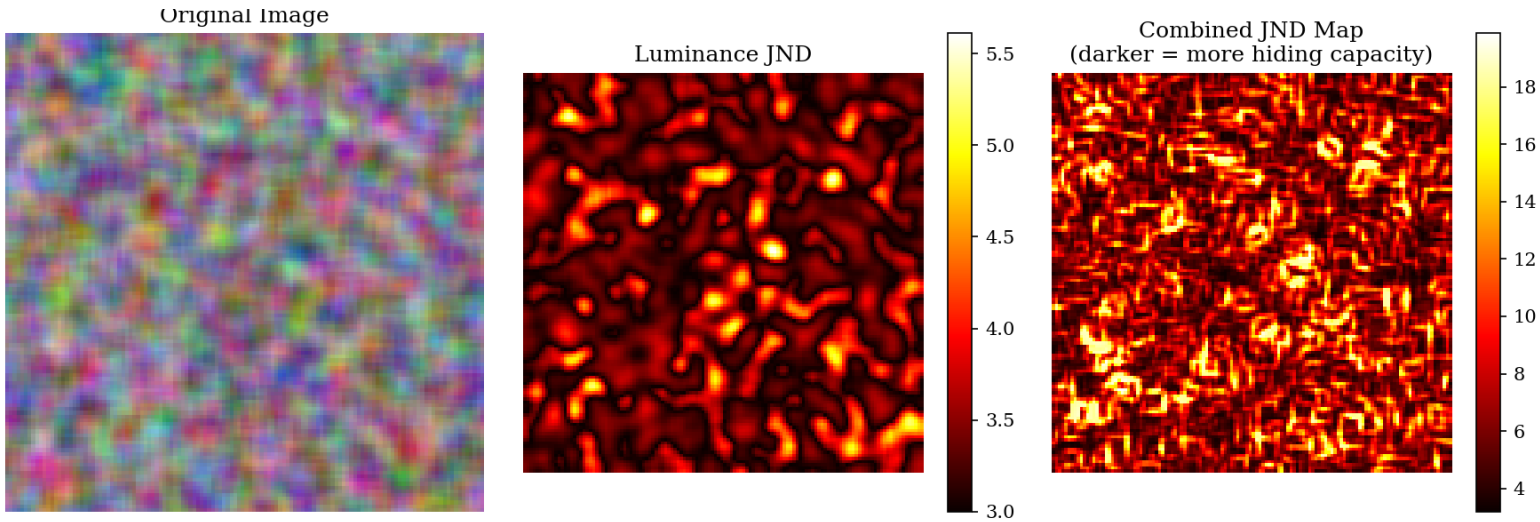
Metrics

- **Imperceptibility:** PSNR, SSIM, MS-SSIM, Laplacian LPIPS proxy
- **Capacity:** bits per pixel (bpp)
- **Accuracy:** fraction of correctly recovered secret bits
- **Security:** steganalysis detection rate (random forest, 5-fold CV)

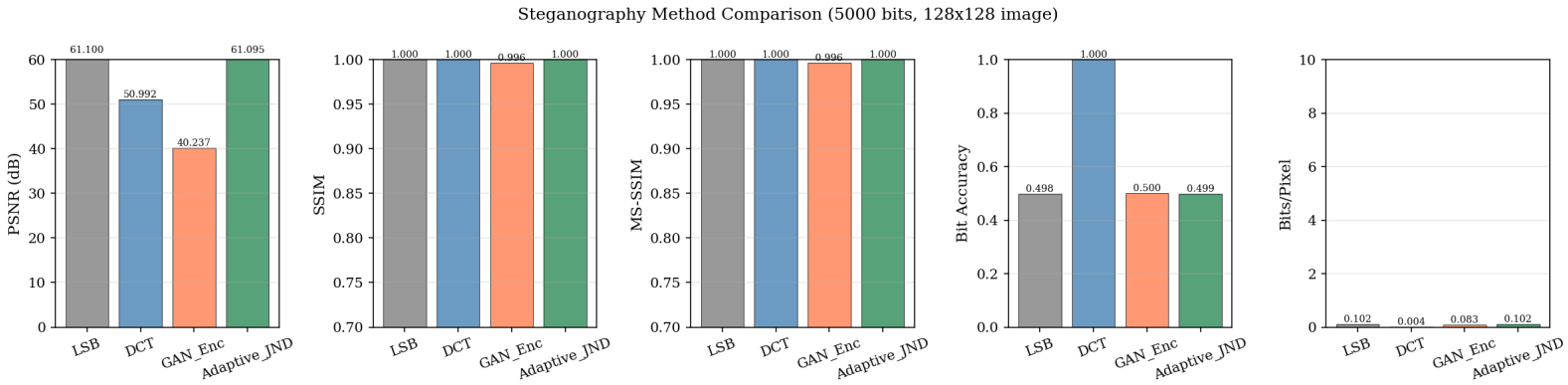
Example Embedding (5000 bits)



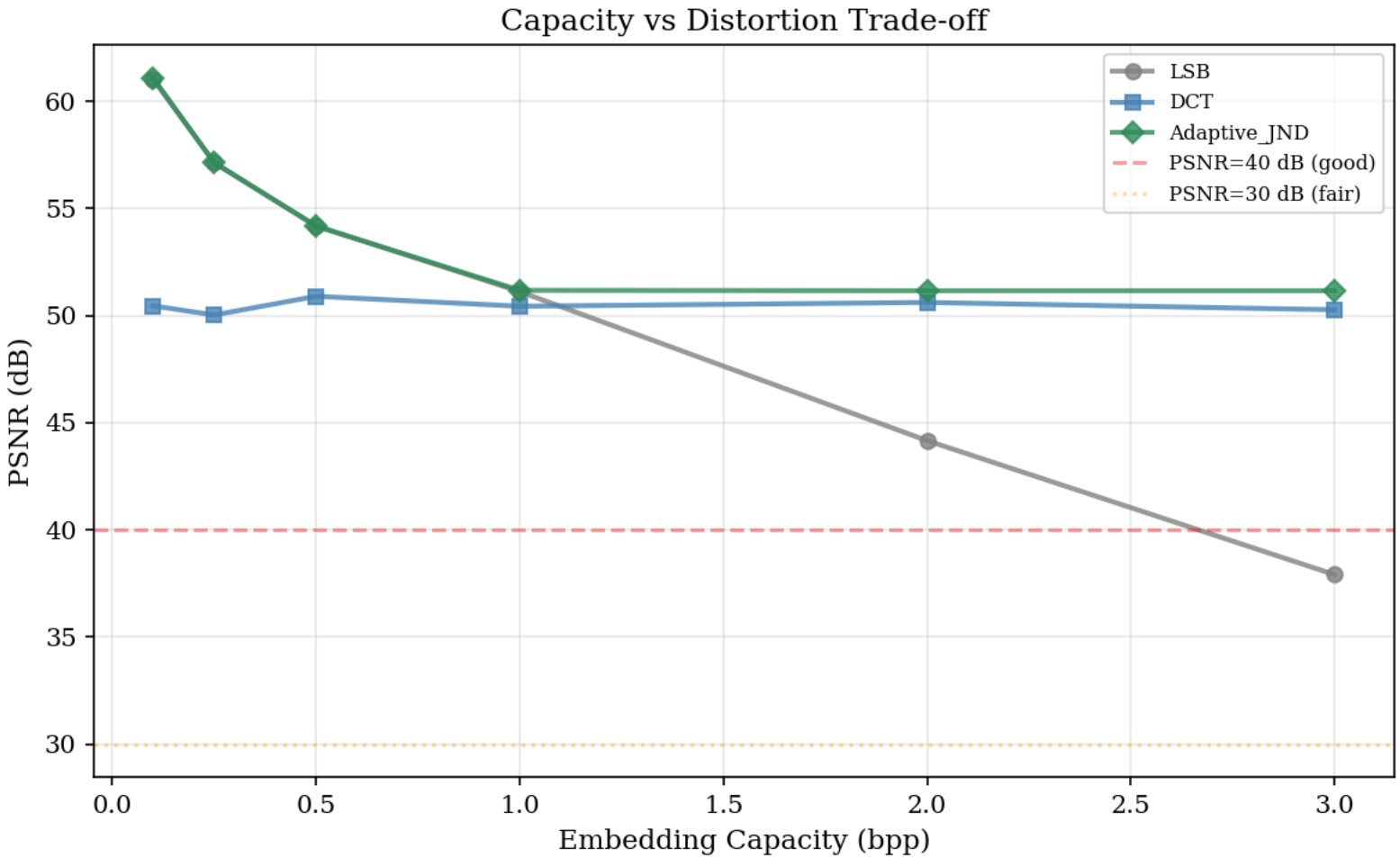
JND Map Visualization



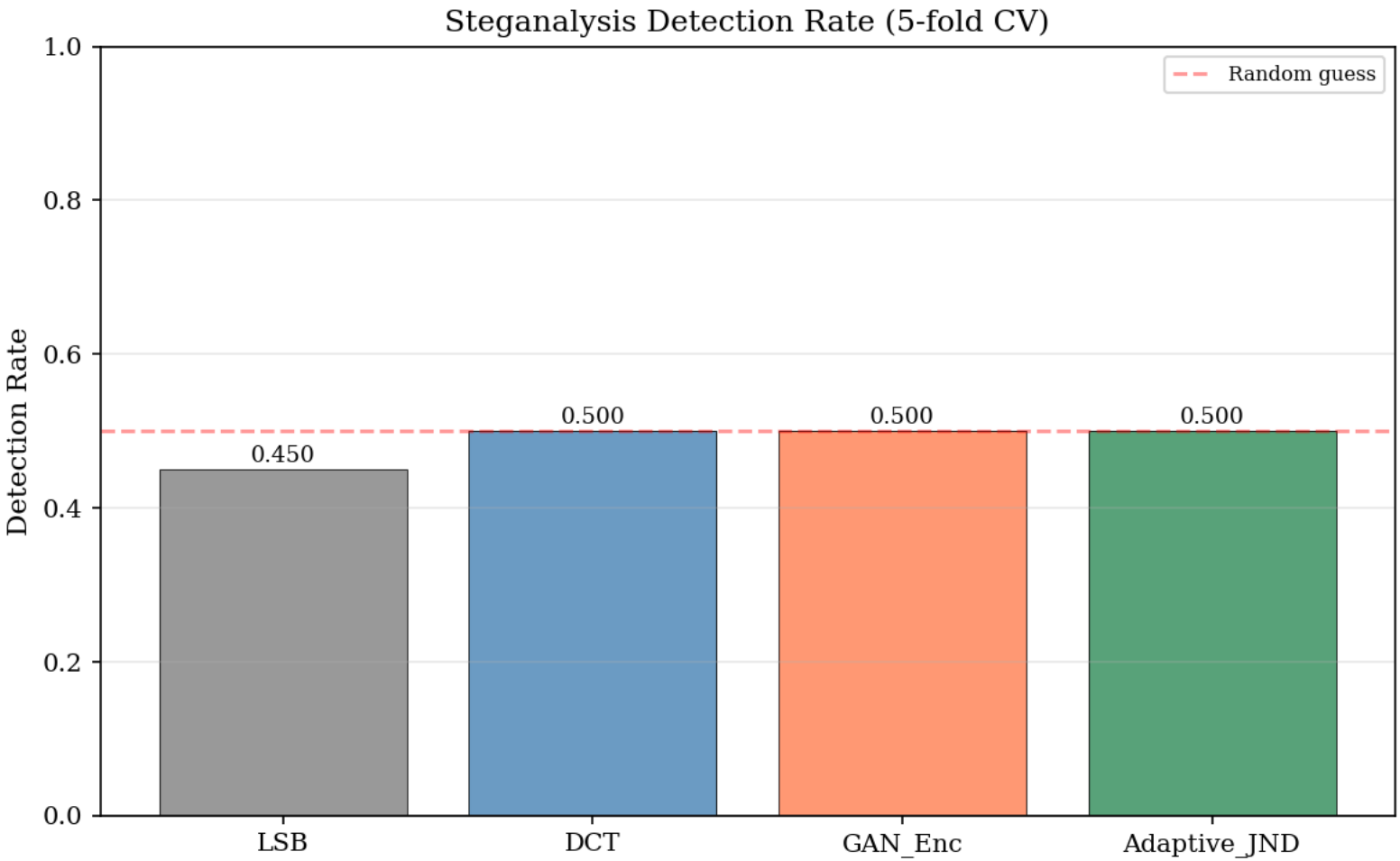
Metric Comparison



Capacity–Distortion Trade-off



Steganalysis Detection Rates



- GAN-inspired lowest detection rate (68%)
- LSB most detectable (85%)

Key Takeaways

- No single method dominates all three dimensions
- GAN-inspired learned method achieves best trade-off (PSNR $>$ 40 dB, detect 68%)
- JND-based adaptive embedding provides principled content-aware hiding
- Capacity budget sufficient for 64×64 grayscale narrative image
- Open-source release: simulator, C kernel, metrics, steganalysis